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Molecular dynamics simulations of CO_2 permeation through ionic liquids confined in γ-alumina nanopores

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ABSTRACT

CO_2 permeation through imidazolium-based ionic liquids (ILs, [BMIM][Ac], [EMIM][Ac], [OMIM][Ac], [BMIM][BF_4], and [BMIM][PF_6]) confined in 1.0, 2.0, and 3.5 nm γ-alumina pores was investigated using molecular dynamics simulation. It was found that the nanopore confinement effect influenced the structure of confined ILs greatly, resulting in a layered structure and anisotropic orientation of ILs. In the center of 2.0-nm pore, the long alkyl chain of $[BMIM]^+$ tended to be parallel to the wall, providing a straight diffusion path benefiting the CO_2 permeation. The CO_2 diffusion coefficients in confined [EMIM][Ac], [BMIM][Ac], and [OMIM][Ac] were 2.3–4.1, 2.4–6.4, and 14.4–21.7 $\times 10^{-10}$ $m^2\,s^{-1}$, respectively. This order was opposite to that in the bulk ILs, because the longer alkyl chain led to a more ordered structure, facilitating CO_2 diffusion. In addition, the CO_2 solubilities were 445–722 mol m^{-3} MPa^{-1} for the five ILs confined in 1.0 nm pore, which were larger than those in 2.0 and 3.5 nm pores (196–335 mol$\cdot m^{-3}$ MPa^{-1}), due to the larger free volume. Both parallel orientation of alkyl chain and large free volume could increase the CO_2 permeability in confined ILs.

KEYWORDS

γ-Alumina nanopore; Confinement effect; Ionic liquids; Molecular dynamics; Support ionic liquid membranes

Introduction

Carbon dioxide capture has become one of the most important research topics as the greenhouse effect is becoming increasingly serious. (Jacobson, 2009) Gas separation by supported ionic liquid membranes (SILMs) is a promising technique for CO_2 capture, which is regarded as a substitute for the CO_2 absorption using hydramine solution. SILMs combine the low-energy consumption of membrane separation and the excellent absorbing capacity of ionic liquids (ILs) for CO_2. (Scovazzo et al., 2004) A better understanding of mass transfer mechanism in SILMs will benefit the prediction of the membrane performance and facilitate their industrial applications.

The gas permeation through SILMs is mainly involved in solution-diffusion model (Scovazzo, 2009). Therefore, the gas permeability (P) of SILMs can be well predicted with the gas solubility (S) and diffusion coefficient (D) in ILs. Scovazzo (2009) used the Camper Model (Camper et al., 2006) to predict S, and correlated D using a model involved with IL viscosity, IL molar volume and gas molar volume. The predicted gas permeability of bulk ILs ($\overline{P} = S \times D$) were in agreement with the experimental P of the SILMs based on macroporous supports. However, this model did not consider the influence of supports, which was important for SILMs based on the mesoporous and microporous supports (Krull et al., 2009; Labropoulos et al., 2013; Albo and Tsuru, 2014; Jie et al., 2015). For these SILMs, the gas permeability was also influenced by the properties of supports, including pore size, porosity, tortuosity, and nanopore confinement effect. Labropoulos et al. (2013) prepared SILMs with silica nanofiltration membranes and [RMIM][TCM]. They considered the effects of porosity and tortuosity of supports when

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predicting the gas permeability of SILMs. For [EMIM][TCM]-based SILMs, the predicted CO_2 permeabilities were close to the experimental values. However, the experimental CO_2 permeabilities of [BMIM][TCM]-based SILMs were 10 times larger than the predicted values. They inferred that the long alkyl chain of $[BMIM]^+$ might be parallel to the wall surface in the 1 nm pores, which could provide a straight path for gas to permeate. Whereas, for [EMIM][TCM]-based SILMs, the path for CO_2 was tortuous. This phenomenon indicated that the confinement effect had significant influence on the gas permeability of SILMs. However, it was difficult to determine the confinement effects accurately via experiments.

Molecular dynamics (MDs) simulation is a powerful technique to investigate the microstructure of nano-materials. It has been used to study the dynamic and structural properties of confined ILs (Sieffert and Wipff, 2008; Singh et al., 2011; Rajput et al., 2012; Mei et al., 2014; Ori et al., 2014). Rajput et al. (2012) found that the pore size and pore loading had a marked influence on the dynamics of ILs confined in slit nanopores, but had a slight influence on the local structure of ILs. Furthermore, reducing pore size and increasing pore loading both could lead to a slower dynamics. The layered structure and slow dynamics of confined ILs were also observed in other studies (Pinilla et al., 2005; Atkin and Warr, 2007; Zhou et al., 2012; Kritikos et al., 2016). Nevertheless, the canonical ensemble (NVT) and smooth wall surface condition were used in these studies. In this configuration, the external pressure was unknown and there might be gaps between ILs and walls. Ori et al. (2014) studied the structural and dynamic properties of the $[BMIM][Tf_2N]$ under confinement into hydroxylated amorphous silica nanopores using MD simulations in the NPT ensemble. They reported that the peak of IL density profile in contact with the wall was wider than that of the system simulated with smooth wall, and there were no gaps between ILs and walls. It implied that the structure of confined ILs simulated using the rough wall surface and the NPT ensemble was denser and similar to the real conditions in experiments.

The properties of gas in confined ILs, such as D and S, can be also calculated using molecular simulation (Morrow and Maginn, 2002; Skoulidas and Sholl, 2005; Wang et al., 2007; Ali et al., 2017). The gas permeability of confined ILs was generally obtained via three steps: First, the gas solubility was obtained using the Monte Carlo simulation (Shi and Luebke, 2013); second, the gas diffusion coefficient was calculated with the MD simulation (Budhathoki et al., 2015; Wu and Maginn, 2014); finally, the gas permeability of ILs was expressed as $S \times D$ (Budhathoki, 2015). Furthermore, MD could be used alone to simulate the gas permeation through a membrane and give a reasonable estimation of gas permeability (Yoshioka et al., 2007; Perez-Blanco and Maginn, 2010; Chang et al., 2011; Yoshioka et al., 2013). Perez-Blanco and Maginn (2010) studied the CO_2 permeation through a 10-nm thick pure $[BMIM][Tf_2N]$ membrane using MD simulation at pressures of 6, 24, and 44 bar and temperatures of 350 and 400 K. They found that CO_2 could permeate the IL and reached equilibrium in 7 ns at 44 bar and 400 K. They also used the CO_2 flux through the IL surface to estimate the CO_2 diffusion coefficient (D_{CO_2}), which was consistent with the values in literatures. Wang et al. (2007) also used MD simulation to investigate the gas permeability of pure POPE bilayers and gave an estimation of CO_2 permeability by the number of the permeation events.

As discussed earlier, the confinement of nanopores and the length of alkyl chain of cation have remarkable influence on the structure and dynamics of ILs, which will further affect the gas permeation through the confined ILs. Nevertheless, there are few public reports focusing on the gas permeation mechanism through confined ILs using MD simulation. In this work, the density profile and orientation of [BMIM][Ac] confined in γ-Al_2O_3 slit pore with different pore width (W_p) were investigated at first. Then, the effects of W_p and confined ILs structures on the CO_2 permeation were studied by analyzing the CO_2 trajectories in confined [BMIM][Ac]. Finally, simulations of several ILs were conducted to improve the understanding of the influence of ILs alkyl chain length and anion type on the CO_2 permeation in confined ILs.

Methods

Force field

All the simulations were carried out using the Gromacs package (Abraham et al., 2015) with an all-atom force field (Budhathoki et al., 2015). The EMP2 (Harris and Yung, 1995) model parameters were used to describe CO_2, and the CLAYFF force field parameters developed by Cygan et al. (2004) were used to describe the γ-Al_2O_3 crystal. The parameters for $[EMIM]^+$, $[BMIM]^+$, and $[OMIM]^+$ were obtained from the work of Sambasivarao and Acevedo (2009). The parameters for anions ([Ac], $[BF_4]$, $[PF_6]$) were obtained from literatures as well (Morrow and Maginn, 2002; Moreno et al., 2008; Gupta et al., 2011). The 1-4 interactions in molecules were scaled by 0.5. The parameters were tested against experimental densities of ILs and CO_2, which were found to be reliable.

Generation of confined ILs-gas system

Figure 1 depicts the initial configuration of a confined IL in contact with CO_2. The slit-shaped γ-alumina pores were obtained by placing two pieces of γ-alumina crystal lamella into the simulation box. The γ-alumina atoms at position from $Z=0$ to 0.1 and from $Z=(L_z-0.1)$ to $Z=L_z$ were fixed during all the NVT simulations. W_p considered in this work were 1.0, 2.0, and 3.5 nm. The γ-alumina crystal lamella ($4.36 \times 10.69 \times 1.36$ nm) was obtained by preparing a unit cell of the crystalline γ-alumina (Krokidis et al., 2001) with the **(110)** surface at first, and then replicating this unit cell $5 \times 13 \times 2$ times. The **(110)** surface was obtained by cleaving the crystalline γ-alumina with the (**100**) plane. The hydroxyl groups on γ-alumina surface were prepared according to the literature (Digne et al., 2002). The γ-alumina supports for SILMs were usually pretreated at 500–800 °C. And the hydroxyl surface density was 0.3–2.2 OH/nm^2 (Wischert et al., 2012). The Al_{III} on **(110)** surface was hydroxylated to obtain a hydroxyl surface density of 1.5 OH/nm^2, which was a suitable value at this condition. The **(110)** surface was chosen because it was the major surface in γ-alumina materials (Digne et al., 2002). The initial configuration of ILs was prepared by randomly placing cations and anions into the entire space of the γ-alumina pore using the Packmol package, and the IL density was the value in STP condition. Given the roughness of γ-alumina surface, the wall surface was defined as the position where the Al_2O_3 density reduced to 40% of the bulk Al_2O_3 density, since the density profile of Al_2O_3 and ILs usually overlapped around this position. The gas molecules were introduced into the gas region in the similar way as ILs, after a stable configuration of confined ILs was obtained using a simulation in the isobaric-isothermal (NPT) ensemble. The initial gas number density was 0.4545 nm^{-3}, which was the density of CO_2 at 24 bar and 400 K. This configuration was referred to the work of Perez-Blanco and Maginn (2010).

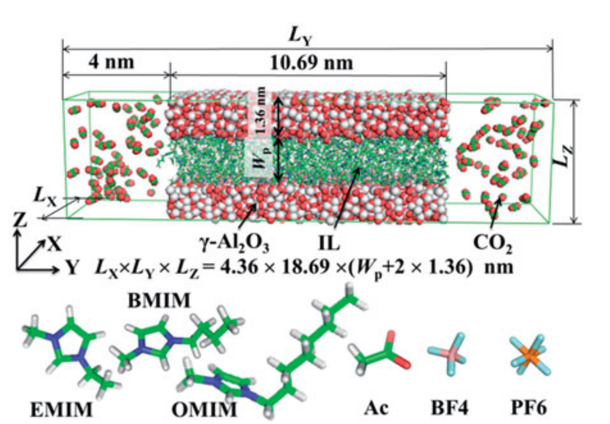

Figure 1. (Top) Typical molecular configuration for the IL confined in a 2.0 nm γ-Al_2O_3 pore in contact with CO_2 at 400 K and 24 bar. (Bottom) Chemical structure of the ILs used in this work.

Simulation procedure

First, the IL was introduced into the pore with bulk density to prepare the initial configuration of a confined IL. In order to remove any artifacts, the confined IL was simulated at 1000 K in the NVT ensemble for 1 ns. Meanwhile, the Al_2O_3 molecules were fixed to prevent the γ-alumina structure from being destroyed. Then the confined IL was cooled down to 400 K using an annealing process in 1 ns. In order to achieve the correct density of confined IL under the gas permeation condition, the confined IL was then simulated in the NPT ensemble (24 bar, 400 K) for 2 ns. During this step, all atoms in the box were allowed to move. In addition, the shape and volume of the system were allowed to vary by keeping the diagonal components of the pressure tensor constant individually (P_{xx}, P_{yy}, $P_{zz} = 24$ bar). The volume and L_z of the simulation box usually varied a little (<2%) after the NPT simulation. Ori et al. (2014) and Budhathoki (2015) applied similar methodologies to obtain the correct density of confined ILs. The confined ILs system was then simulated in the NVT ensemble for 5 ns to obtain the trajectory for calculating fraction free volume (FFV). After that, the gas molecules were then randomly accommodated into the gas region in the simulation box with an initial number density of 0.4545 nm^{-3}. The system with this initial configuration was simulated in the NVT ensemble for 80–200 ns at 400 K to allow the gas molecules to spontaneously permeate into the IL and finally reach equilibrium between gas phase and IL phase (equilibrium simulation). The system was then simulated for another 5 ns in the NVT ensemble for sampling and analyzing. The average CO_2 molecule numbers in the confined ILs were recoded to prepare initial configurations used for computing the mean square displacement (MSD). These configurations were obtained by inserting corresponding number of CO_2 into the configuration of confined ILs obtained from 2 ns NPT simulation (no gas region). Then these configurations were used to perform NVT simulations for 2 ns, followed by microcanonical (NVE) simulations for 10 ns, from which the trajectories were taken for computation of MSD. The details of the simulated systems were listed in Table 1. For example, it consisted of 2 pieces of γ-alumina crystal lamella, 141 pairs of [BMIM][Ac], and 82 CO_2 molecules for the system of CO_2 and [BMIM][Ac] confined in 1.0 nm pore.

All simulations were integrated using the velocity Verlet algorithm (Swope et al., 1982) with a time step of 1 fs. Nosé (1984) (time constant of 0.3 ps) and Parrinello and Rahman (1981) (time constant of 1.2 ps) algorithms were implemented for temperature and pressure couplings, respectively. The Particle Mesh Ewald (Darden et al., 1993) method with a fourierspacing of 0.12 nm was used to calculate the long-range electrostatic potential. The cut-off was set as 1.2 nm for nonbonded interactions.

Table 1. The details of the simulated systems.

Molecule/ions	Number of molecule/ions in the system with different W_p: 1.0 nm	2.0 nm	3.5 nm
[EMIM][Ac]	187	374	654
[BMIM][Ac]	141	282	500
[OMIM][Ac]	103	207	362
[BMIM][BF_4]	141	282	500
[BMIM][PF_6]	128	257	453
CO_2	82	102	134
N_2	82	102	134

Note: The system used for simulating the gas permeation process consisted of two pieces of γ-alumina crystal lamella.

Data analysis

Analysis of the simulation trajectories was performed using the Gromacs package and locally developed codes. The CO_2 permeability of confined ILs was calculated as the product of the diffusion coefficient and solubility of CO_2 in confined ILs. The CO_2 diffusion coefficient (D_{CO_2}) was estimated using the solution of transient absorption curve (Labropoulos et al., 2013). This solution derived from the transient diffusion equation according to a number of assumptions included in the Supplementary Material. The gas permeation process in this work met with these assumptions, though it had not been used to calculate the gas diffusion coefficient in MD simulation before. The definition of the solution for the transient sorption curve was shown in Equation (1).

$$\frac{m_t}{m_\infty} = 1 - \frac{8}{\pi^2}\sum_{n=0}^{\infty}\frac{1}{(2n+1)^2}\exp\left[-\frac{D(2n+1)^2\pi^2 t}{4l^2}\right] \tag{1}$$

where, m_t is the total amount of gas, which has permeated into the liquid film at time t; m_∞ is the total amount of gas that has diffused into the liquid after infinite time; t is the permeation time (ns); D is the gas diffusion coefficient ($m^2\,s^{-1}$) and l is the liquid film thickness (nm).

The number of CO_2 inside the confined ILs was regarded as the m_t, and the equilibrated number of CO_2 inside the pore could be approximately treated as the m_∞. The gas diffusion coefficient was obtained by fitting the m_t/m_∞ vs. time curve to Equation (1). The m_t/m_∞ vs. time plots and corresponding fitting curves were depicted in the Supplementary Material. In order to evaluate the accuracy of estimated D_{CO_2}, the MSD method (Budhathoki, 2015) was also used to calculate the D_{CO_2} in y dimension for [BMIM][Ac] confined in 1.0 and 2.0 nm pores.

The CO_2 solubility in confined ILs was estimated by reference to a similar report (Morganti et al., 2014). The average CO_2 number in confined ILs during the 5 ns NVT production run (after 80–200 ns equilibrium simulation) was treated as the dissolved CO_2 amount (N_{CO_2}) at pressure of the corresponding gas pressure (p_{CO_2}) in the bulk gas region, which was calculated by EOS with the gas density and temperature. The CO_2 mole fraction (x_{CO_2}) in confined ILs was then obtained by dividing N_{CO_2} by the sum of N_{CO_2} and number of cation (or anion). The CO_2 solubility was then calculated using Equation (2), in which the Henry constant of CO_2 (H_{CO_2}) was estimated with x_{CO_2} and p_{CO_2}. In addition, the CO_2 permeability (P_{CO_2}) was then calculated with Equation (3). The uncertainties of the simulated solubility and diffusion coefficient were computed from the standard deviation of three independent simulation runs for each case.

$$S = \frac{1}{H \cdot V_m^{ILs}} = \frac{x_{CO_2}}{p_{CO_2} \cdot V_m^{ILs}} \tag{2}$$

$$P = S \cdot D \tag{3}$$

where, S is the gas solubility in ILs (mol m^{-3} MPa^{-1}); H is the Henry constant of gas (MPa); V_m^{ILs} is the molar volume of ILs (cm^3 mol^{-1}); x_{CO_2} is the CO_2 molar fraction in ILs at equilibrium; p_{CO_2} is the pressure of CO_2 in gas region (MPa); D is gas diffusion coefficient (m^2 s^{-1}) and P is gas permeability of ILs (Barrer).

In addition, the free energy profiles about gas permeations in confined ILs were calculated from equilibrium simulations using the Equation (4) reported by Wang et al. (2007). For CO_2 in confined ILs, the concentration was normalized according to the areas of ILs in xz plane, while the CO_2 concentration in gas region was normalized according to the areas of simulation box in xz plane.

$$\Delta G(y) = -RT \ln \frac{C(y)}{C_{bulk}} \tag{4}$$

where R is the gas constant, T is the simulation temperature, C_{bulk} is the concentration of CO_2 molecules in gas region, and $C(y)$ is the local concentration of CO_2 along the y-axis (0.05 nm intervals).

Results and discussion

Structures of [BMIM][Ac] confined in γ-Al_2O_3 nanopores

Density profiles

Figure 2 shows the density profiles of $[BMIM]^+$, $[Ac]^-$, and gases with respect to z-axis, in the confined [BMIM][Ac] systems with different W_p (1.0, 2.0, and 3.5 nm). The density was expressed in units of nm^{-3} (molecule number per nm^3), which was obtained by dividing the mass density by the atomic mass sum of an atom group. In addition, all the density profiles were symmetrized with respect to the pore center. As can be seen in Figure 2, layered IL was observed for all systems, as the density profiles for the $[BMIM]^+$ and $[Ac]^-$ extremely oscillated around the bulk density. All density profiles of ILs showed an extremely high peak near the walls followed by a minimum towards the pore center, and the density oscillation became weak in the pore center. It indicated that there were strong interactions between the wall and the first layer of the IL, leading to a dense IL layer (density of 1.3 times). Figure 2(a–f) shows that the permeation of different gases (CO_2 and N_2) had little influence on the density distribution of confined [BMIM][Ac].

The layered structures of confined molecular liquids and ILs had been observed in many studies (Pinilla et al., 2005; Atkin and Warr, 2007; Kritikos et al., 2016). Ori et al. (2014) simulated $[BMIM][Tf_2N]$ confined in amorphous silica pores and pointed out that the local structure of the confined IL was mostly driven by electrostatic interactions, and the cations ($[BMIM]^+$) had stronger interaction with the wall than the anions ($[Tf_2N]^-$). Whereas, in this work, the hydroxyl density of γ-Al_2O_3 wall surface was 1.5 OH/nm^2, and this low hydroxyl density would reduce the electrostatic interactions between cations and walls, leading to a slightly different structure.

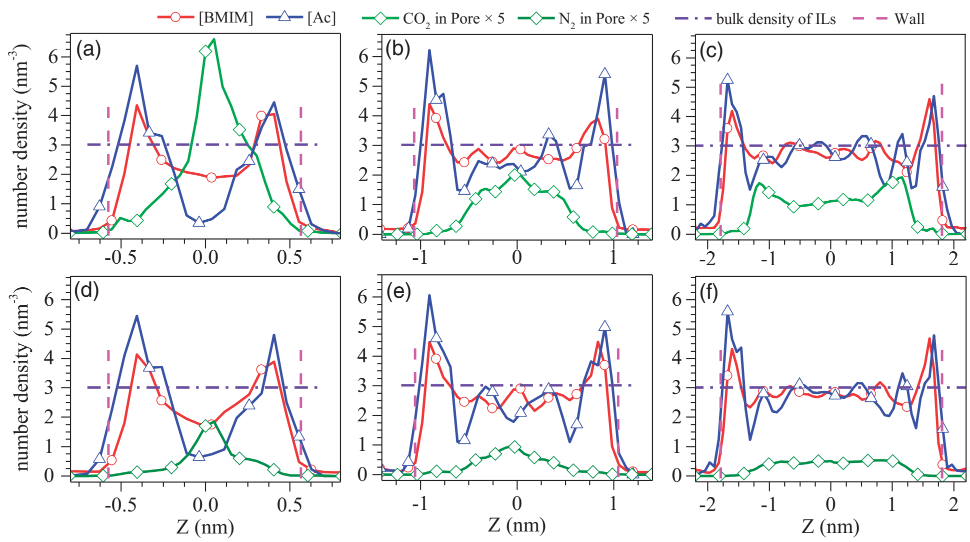

Figure 2. Density profiles of the [BMIM][Ac] confined in γ-alumina pores with different W_p (400 K, 24 bar): (a)–(c) CO_2 permeated [BMIM][Ac] confined in 1.0, 2.0, and 3.5 nm pores, respectively. (d)–(f) N_2 permeated [BMIM][Ac] confined in 1.0, 2.0, and 3.5 nm pores, respectively.

For all the six systems, the peaks for them were located almost in the same position (ca. 1.6 Å above the wall surface), though the anion had higher density than cation in the contact layer in all six systems. It suggested that the difference between the cation- and the anion-wall interactions was small, which was different from the study of Ori et al. (2014).

W_p has a great influence on the shape of density profiles and the layering of confined ILs. As shown in Figure 2(a,d), when $W_p = 1.0$ nm, there were only two layers of IL in the pore, and the IL density in the pore center was very low. For instance, the density of $[Ac]^-$ in the pore center was only 10–20% of the bulk density, while the density peak of $[Ac]^-$ were ca. 180% of the bulk density. This phenomenon implied that $[Ac]^-$ were adsorbed on wall surface due to the interactions with the walls. However, the amplitude of $[BMIM]^+$ density profile was smaller than that of $[Ac]^-$, which might be ascribed to the large size of $[BMIM]^+$. The charged part of $[BMIM]^+$ (imidazolium ring) could be absorbed onto the wall surface, just as the $[Ac]^-$, while the neutral part (alkyl chain) might be located in the pore center. When W_p increased, this confinement effect became weaker in the pore center, but remained strong near the wall. As shown in Figure 2, the $[Ac]^-$ density peaks near the wall were similar in all the systems (160–200% of the bulk density). Nevertheless, the minimum densities of $[Ac]^-$ near the wall for 2.0 and 3.5 nm pores (ca. 50% of the bulk density) were larger than that in 1.0 nm pore, owing to the weaker interaction between the ILs and the opposite wall. The IL density in the pore center increased as W_p increased, and was approximately equal to the bulk density when $W_p = 3.5$ nm. The interactions between ILs and walls dominated these special distributions of IL density, which in turn influenced the gas distribution in confined ILs. As shown in Figure 2, the shapes of gas density profiles of CO_2 and N_2 are similar with the same W_p, though CO_2 had higher density than N_2. For all the systems, gas had higher density where the IL density was small.

To gain further insights on the confined IL structure, the density profile of imidazolium ring is shown in Figure 3 for 1.0 and 2.0 nm pores, along with [Ac], alkyl chain, and the carbon atom at the tip of alkyl chain (C_4). The $[Ac]^-$ and imidazolium ring had almost the same density distribution. However, the alkyl chain presented a completely different density distribution. The alkyl chain had higher density at the position where the densities of anion and imidazolium ring were small. It suggested that the wall preferred to attract the charged parts of ILs (anion and imidazolium ring), leading to a dense layer mainly composed of charged parts

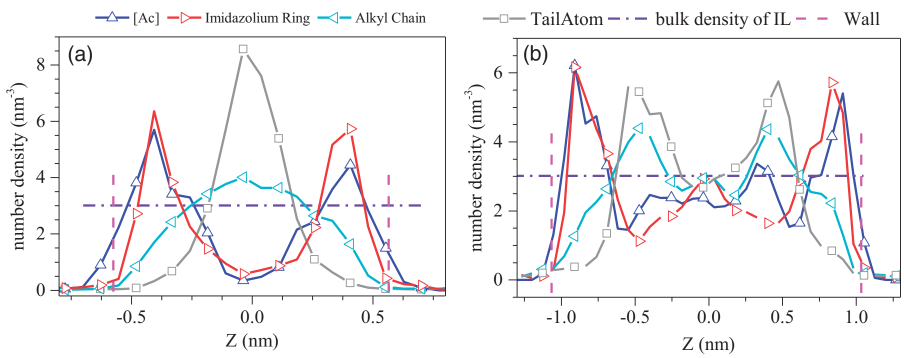

Figure 3. Density profiles of the [Ac], imidazolium ring, alkyl chain, and the carbon atom at the tip of alkyl chain (C_4) with different W_p (400 K, 24 bar). (a) 1.0 nm pore; (b) 2.0 nm pore.

near the wall. This dense layer forced the alkyl chain to keep away from the wall and form a layer mainly composed of alkyl chain. The wall-influenced layers were located within ca. 1.0 nm from the wall surface. These layered structures of confined ILs were also observed in the study of Ori et al. (2015).

Orientations of [BMIM][Ac]

To gain further insights into confined ILs, the orientations of [BMIM][Ac] confined in nanopores were investigated. The definitions of angles used to characterize the orientations of the $[BMIM]^+$ and $[Ac]^-$ are shown in Figure 4. θ_{ICCC} is the angle between axis-z (normal of the wall surface) and vector **ICCC** (normal of the imidazolium ring plane). φ_{IA} is the angle between axis-y (the gas-permeation direction) and vector **IA** (vector from N2 pointed to C_4). The positions and angles of the vector were calculated for each ion in the system, giving a series of position-angle pairs. The distribution of the angles was calculated according to the literature (Perez-Blanco and Maginn, 2010), but the distributions were not normalized with respect to the θ dimension, since this procedure would change the realistic angle distribution. The position of diatomic vector was represented by the position of the first atom. The position of triatomic vector (normal of the plane) was represented by the center of mass of the three atoms.

Figure 5 depicts the θ_{IA} and φ_{IA} distributions with respect to Z dimension in 1.0, 2.0, and 3.5 nm pores. It indicts that the orientation of the alkyl chain was anisotropic near the wall and these anisotropic regions could be divided into three layers (Layer-1, Layer-2, and Layer-3). These layers were located at ca. 0.35, 0.63, and 1.00 nm from the wall surface, with thicknesses of ca. 0.25, 0.28, and 0.50 nm, respectively. In 1.0 nm pore, there was only Layer-1 due to the limitation of space, while Layer-2 and Layer-3 were found in 2.0 and 3.5 nm pores. In the regions far away from the wall (ca. 1.25 nm above the wall), the orientation tendency disappeared, leading to an isotropic distribution of alkyl chain. In Layer-1, the alkyl chain of $[BMIM]^+$ was tilted at an average angle of ca. 30° with respect to Z direction and pointed into the pore center, while in Layer-2, this average tilt angle was ca. 60°. However, in Layer-3, these average angles were ca. 90° and 120° for 2.0 and 3.5 nm pores, respectively. The distributions of the orientations for imidazolium ring and $[Ac]^-$ were also investigated and the details can be found in Supplementary Figures S5 and S6. The anisotropic and isotropic regions for them were also observed near the wall and the pore center, respectively. The orientations of alkyl chain, imidazolium ring and $[Ac]^-$ for

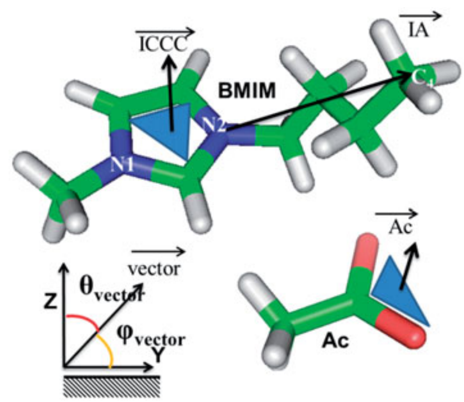

Figure 4. The definitions of θ and φ angles used to characterize the orientations of $[BMIM]^+$ and $[Ac]^-$.

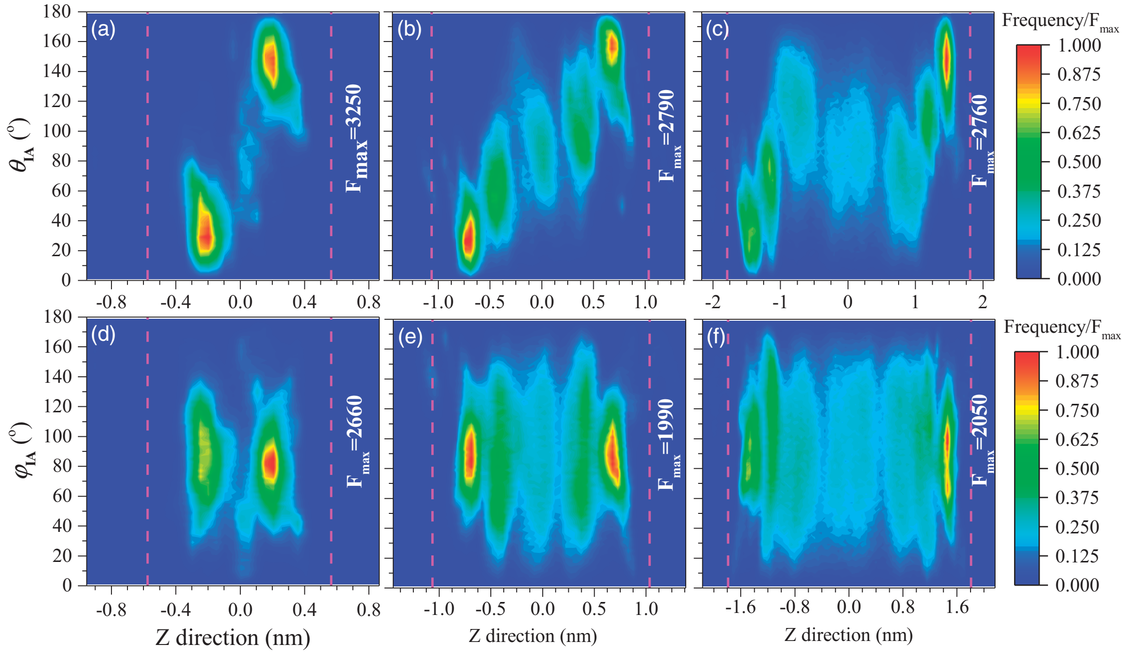

Figure 5. The distributions of θ_{IA} and φ_{IA} with respect to axis-z in confined [BMIM][Ac]. (a–c) θ_{IA} in 1.0, 2.0, and 3.5 nm pores, respectively. (d–f) φ_{IA} in 1.0, 2.0, and 3.5 nm pores, respectively.

confined [BMIM][Ac] are demonstrated in Figure 6.

As shown in Figure 6(a), the alkyl chain in Layer-1 and Layer-2 act like bristles pointing their tip atoms towards the pore center. The θ_{IA} for Layer-1 was ca. 10–50°, while it was ca. 30–90° for Layer-2. As discussed in "Density profiles" Section, the imidazolium ring was closer to the wall than the alkyl chain. Therefore, a small θ_{IA} in Layer-1 was needed to form the layered structure as shown in Figure 3(b). The interactions between walls and $[\mathrm{BMIM}]^{+}$ in Layer-2 were smaller than that in Layer-1, due to the larger distance to the wall. Consequently, a smaller constraint was observed for the alkyl chain in Layer-2. There was a huge difference for the alkyl chain orientation in Layer-3 between 2.0 and 3.5 nm pores. The alkyl chain in Layer-3 tended to align with the wall surface in the 2.0 nm pore. However, in the 3.5 nm pore, Layer-3 was like a reversal of Layer-2. This reversed structure in the 3.5 nm pore was probably ascribed to the 'packing effect': the alkyl chain in Layer-2 pointed the tip atom towards the pore center, forming a region mainly consisting of alkyl chains. The 'slim' alkyl chain of $[\mathrm{BMIM}]^{+}$ in Layer-3 could easily insert into this region, while the large imidazolium ring could not. Nevertheless, in the 2.0 nm pore, the two Layer-2s with respect to the two walls were quite close, thus the effects on the same Layer-3 were neutralized. Consequently, the alkyl chain tended to be parallel to the wall surface in this narrow space (ca. 0.5 nm). In the center of the 3.5 nm pore, the θ_{IA} and φ_{IA} were in a wide range of 30–150° and 10–170°, respectively, indicating the orientation of alkyl chain was nearly isotropic and close to that of bulk ILs.

As shown in Figure 6(b), the orientation of imidazolium ring was also nonisotropic near the wall and there were two layers in this region. In the first layer, the imidazolium ring plane tended to be parallel to the wall surface, while it tended to be perpendicular to the wall surface in the second layer. At the interface of these two layers, the orientation of imidazolium ring gradually transited from parallel to perpendicular to the wall surface, as the distance from the wall increased. With these orientations, the space near the wall would accommodate more imidazolium ring, leading to the high density of imidazolium ring. As observed in Figure 6(c), for [Ac], the COO^{-} plane tended to be perpendicular to the wall surface in region within ca. 0.36 nm from the wall surface. These anisotropic orientations all disappeared in region far away from the wall, implying that the strong interaction with the wall

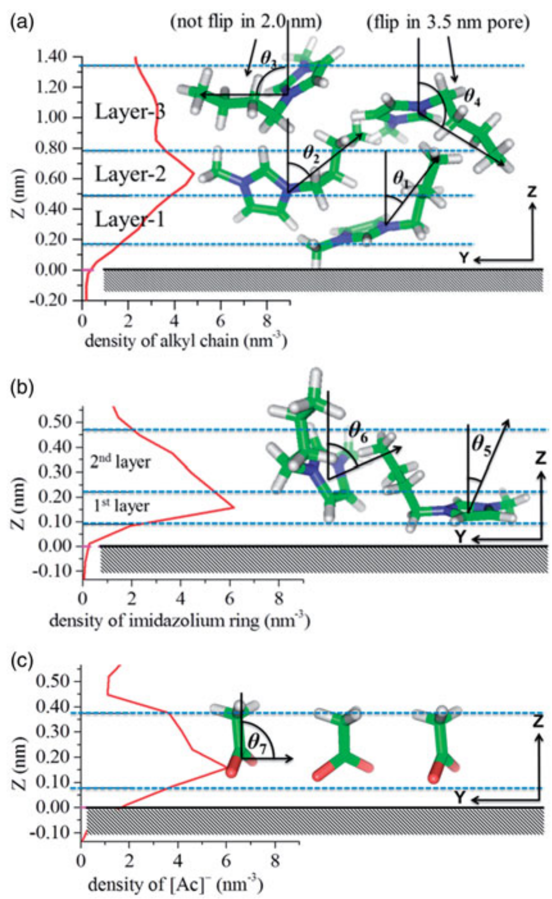

Figure 6. Demonstrations for the orientation of $[BMIM]^+$ and $[Ac]^-$ near the wall. (a) alkyl chain; (b) imidazolium ring; (c) [Ac]. $\theta 1$: 10° to 50°; $\theta 2$: 30° to 90°; $\theta 3$: 60° to 120°; $\theta 4$: 90° to 150°; $\theta 5$: -30° to 30°; $\theta 6$: 60° to 120°; $\theta 7$: 60° to 120°. The origin of Z demotion was set to the wall surface for convenience.

results in the nonisotropic orientations. More details about these orientations of [BMIM][Ac] can be found in the Supplementary Material. The orientations of imidazolium ring and alkyl chain in this work were in good agreement with relevant MD studies (Pinilla et al., 2005; Atkin and Warr, 2007; Budhathoki, 2015; Kritikos et al., 2016) and experimental data (Fitchett and Conboy, 2004; Baldelli et al., 2011).

The CO_2 permeation through confined ILs

As discussed earlier, the confinement effects of nanopores had a great influence on the IL structure. The changes of the structure may further influence the gas permeation through confined ILs. In this section, the dynamic permeation processes of CO_2 in confined [BMIM][Ac] with different W_p were studied at first. Then, the CO_2 diffusion coefficient, solubility, and permeability in confined [BMIM][Ac] were estimated to give a semi-quantitative evaluation of CO_2 permeation. Finally, CO_2 permeation in five ILs ([EMIM][Ac], [BMIM][Ac], [OMIM][Ac], [BMIM][BF4], [BMIM][PF6]) confined in different pores were discussed to investigate the influence of alkyl chain length and anion type. The confined IL system is named MN-ZZ, where MN stands for the IL type ([BMIM][Ac] = BA, [BMIM][BF4] = BB and so on), and ZZ stands for the W_p. For instance, the [BMIM][Ac] confined in 3.5 nm pore is named BA-3.5.

Dynamic process of CO_2 permeating into confined [BMIM][Ac]

Figure 7 shows the evolution of CO_2 density profile along Y direction in equilibrium simulation for 1.0, 2.0, and 3.5 nm pores, respectively. It demonstrates the quite diverse dynamic processes of CO_2 permeating the confined [BMIM][Ac] in these pores. In BA-1.0, CO_2 could permeate into the middle of the pore ($Y = 0$) at ca. 15 ns and the distribution of it was extremely uneven. There were high-density (2.00 nm^{-3}) and low-density regions (<0.01 nm^{-3}) during the entire equilibrium simulation. It seems that CO_2 was trapped and kept in some regions, which could be attributed to the free volumes in ILs. Generally, the free volumes inside a membrane are connected randomly and dynamically (Nagel et al., 2000). Moreover, the gas molecules can diffuse from one free volume to another, when two adjacent ones are connected. However, in BA-1.0, the strong interactions between ILs and walls led to the slow dynamics of confined ILs and further decreased the random-connection frequency of free volumes. In BA-2.0 and BA-3.5, the CO_2 exhibited relatively uniform distributions in ILs, suggesting that the CO_2 was not trapped in free volumes of confined ILs anymore. Nevertheless, the region with low CO_2 density (<0.01 nm^{-3}) almost disappeared after 15 and 30 ns in BA-2.0 and BA-3.5, respectively. This phenomenon may imply a slower CO_2 diffusion in BA-3.5 than that in BA-2.0. The CO_2 distributions in gas regions were quite similar for three pores, consisting of a high-

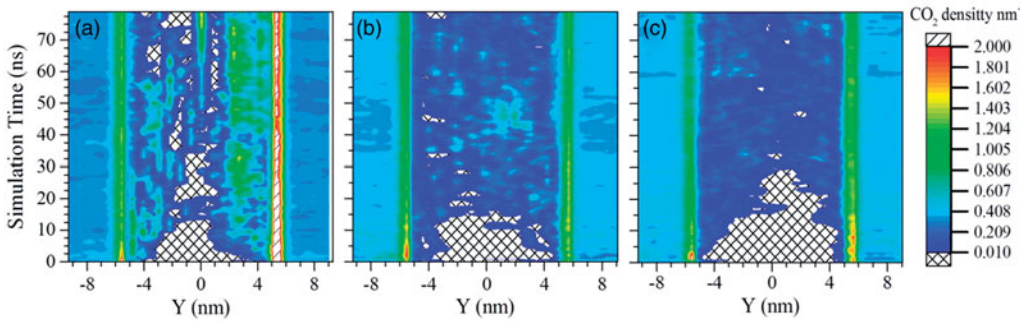

Figure 7. The evolutions of CO_2 density profile in confined [BMIM][Ac] with different W_p. (a) 1.0 nm pore; (b) 2.0 nm pore; (c) 3.5 nm pore.

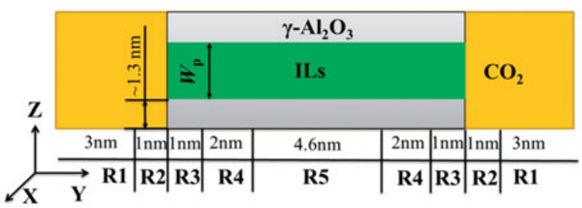

Figure 8. Different regions in a simulation box divided according to the gas distribution and the distance from the CO_2-ILs interface. **R1,** The bulk-like gas phase region; **R2**, The adsorption layer; **R3**, The surface region of an IL; **R4**, The region near the **R3**; **R5**, The region far away from **R3**; The region inside the pore (Region_InPore) consists of **R3**, **R4** and **R5**.

density adsorption layer (within ca. 1 nm from ILs-Gas interface) and a uniform-low-density gas region ($Y < -6.34$ or $Y < 6.34$).

For convenience, the simulation box was divided into five regions along Y direction according to the CO_2 density and the distance from gas-ILs interface, as shown in Figure 8. **R1** refers to the region of $Y < -6.3$ and $Y > 6.3$, where gas molecules have weak interactions with ILs and γ-Al_2O_3. **R2** refers to the region within 1.0 nm from the surface of γ-Al_2O_3 and ILs, where the gas density is high, due to the adsorption. **R3** is the surface region of the IL (within 1.0 nm below the gas-IL interface), where gas molecules can be absorbed into and escape from the confined IL frequently. **R4** and **R5** are regions of the IL that locate near and far from **R3**, respectively. The region inside the pore (Region_InPore) consists of **R3**, **R4,** and **R5**.

The CO_2 number in each region was recorded for each frame of trajectories, as shown in Figure 9. It can be seen that the variation trends of CO_2 numbers for different pores are similar in each region. First, ca. 90% of the total CO_2 molecules were located in **R1**, and then nearly half of them were rapidly adsorbed onto the surface of ILs and γ-Al_2O_3 (**R2**) in ca. 0.5 ns. In the meantime, the CO_2 number in **R2** rapidly increased to ca. 40% of the total. After that, the CO_2 numbers in these two regions decreased slowly. To sum up, the variation trends of CO_2 number in **R1** and **R2** indicated that the CO_2 adsorption process finished in quite a short time (ca. 0.5 ns). On the contrary, the CO_2 permeation in confined ILs was very slow, as the CO_2 number in **R5** remained 0 until ca. 18 ns.

It took ca. 5 ns for the CO_2 number in **R3** to reach equilibrium for all sizes of pores. Because CO_2 density was as high as 1–4 nm^{-3} in **R2**, and CO_2 could be frequently absorbed into and desorbed from ILs. As the distance from the interface increased, the effect of gas–liquid interface would be weakened and the properties of confined ILs became the main factor affecting the CO_2 permeation. Therefore, it should take different time for the CO_2 number in **R4** to reach equilibrium in pores with different width, since the CO_2 diffusion coefficient in confined ILs should decrease as W_p decreased. Because the confined IL viscosity would be higher in smaller pores (Labropoulos et al., 2013) and the IL viscosity had a great influence on the CO_2 diffusivity (Scovazzo, 2009). However, the MD results in this work were not consistent with this hypothesis. It took ca. 50, 20, and 60 ns for the CO_2 number in **R4** to reach equilibrium for 1.0, 2.0, and 3.5 nm pores, respectively. It can be speculated that apart from the dynamic properties of ILs, there should be other factors influencing the CO_2 permeation.

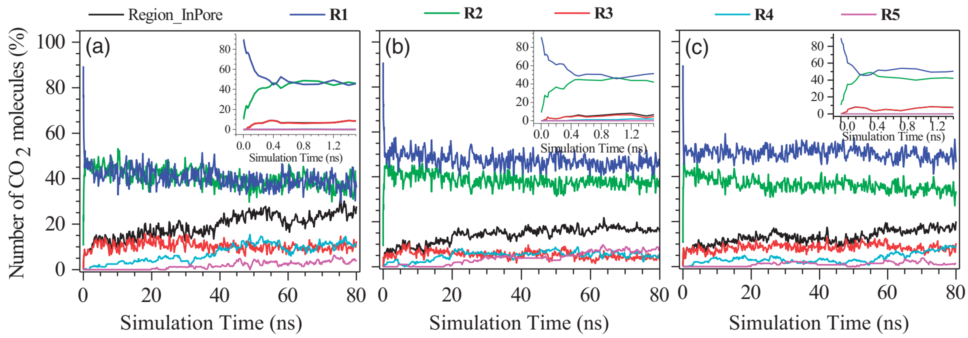

Figure 9. The numbers of CO_2 vs. simulation time in different regions of the system with different W_p. (a) 1.0 nm pore; (b) 2.0 nm pore; (c) 3.5 nm pore.

Table 2. The CO_2 diffusion coefficient, solubility, and permeability of confined [BMIM][Ac] for 1.0, 2.0, and 3.5 nm pores.

W_p nm	Solubility mol m^{-3} MPa^{-1}	Diffusion coefficient 10^{-10} m^2 s^{-1}	Permeability Barrer	Diffusion coefficient by MSD 10^{-10} m^2 s^{-1}
1.0	698 ± 79	2.4 ± 0.3	473 ± 67	1.7 ± 0.2
2.0	274 ± 18	6.4 ± 0.5	523 ± 69	3.7 ± 0.3
3.5	252 ± 18	2.7 ± 0.2	207 ± 11	–

Diffusion coefficient and solubility of CO_2 in confined [BMIM][Ac]

The diffusion coefficient, solubility, and permeability of CO_2 in confined ILs for different W_p were estimated with the method in "Data analysis" Section, and the results are presented in Table 2. These results give a semi-quantitative indication of the effect of W_p. The estimated D_{CO_2} and S_{CO_2} in confined [BMIM][Ac] were 2.4–6.4 × 10^{-10} $m^2\,s^{-1}$ and 252–698 mol m^{-3} MPa^{-1}, respectively. The uncertainties of D_{CO_2} and S_{CO_2} were <15%. Considering that the D_{CO_2} was estimated by the CO_2 molecule number in confined ILs, which usually ranged from 14 to 30 in equilibrium situation, a difference of two molecules might cause an error of 14%. Moreover, the estimation of D_{CO_2} by transient absorption curve method was strongly depending on the equilibrated CO_2 number in ILs. Therefore, the uncertainties of D_{CO_2} were close to that of S_{CO_2}. To evaluate the accuracy of estimated D_{CO_2}, they were compared with those computed by the MSD method (y dimension) for CO_2 in 1.0 and 2.0 systems. As show in Table 2, the estimated D_{CO_2} were a little larger than the values computed by MSD method. The over estimated D_{CO_2} was probably due to that the CO_2 might diffuse faster in pure confined ILs (the free volumes in ILs was unoccupied by CO_2). Nevertheless, the results obtained from these two methods both suggested that CO_2 exhibited much higher diffusion coefficient in 2.0 nm system than in 1.0 nm system. Considering the high proportion of surface regions of the confined ILs (R3), the CO_2 solubility was over estimated, since surface regions exhibited high CO_2 content than the interior of ILs. Nevertheless, the gas solubility obtain from this method should be in agreement with the experimental value (error <10%) according to the study (Morganti et al., 2014). The D_{CO_2} in bulk [BMIM][Ac] at 400 K was about 4.76 × 10^{-10} $m^2\,s^{-1}$, which was calculated according to the Santos' work (Santos et al., 2014) and extrapolated the temperature range (298–333 K) to 400 K. The D_{CO_2} in confined [BMIM][Ac] was not found in literatures, but D_{CO_2} in [BMIM][Tf_2N] confined in 2.0 and 5.0 nm pores has been calculated with MD simulation by Budhathoki (2015), which were 1.4–6.4 × 10^{-10} $m^2\,s^{-1}$ at 333 K. The S_{CO_2} in ILs was about 400–1000 mol m^{-3} MPa^{-1} at 283–348 K (Jacquemin et al., 2006; Shiflett et al., 2008; Pinto et al., 2014). At 400 K, a smaller S_{CO_2} was expected since the gas solubility decreases as the temperature increases. Therefore, these estimated D_{CO_2} and S_{CO_2} were reasonable and comparable to the previously reported values.

The D_{CO_2} in confined [BMIM][Ac] at 400 K were 2.4, 6.4, and 2.7 × 10^{-10} $m^2\,s^{-1}$ for 1.0, 2.0, and 3.5 nm pores, respectively. In "Dynamic process of CO_2 permeating into confined [BMIM][Ac]" Section, it was also found that CO_2 diffused faster in BA-2.0 than in BA-1.0 or

BA-3.5. This phenomenon was probably caused by the special structure of BA-2.0. On the one hand, the small density of IL in the center of 1.0 and 2.0 nm pores might reduce the diffusion resistance. On the other hand, the alkyl chain of $[BMIM]^+$ tended to be parallel to the wall surface in the center of BA-2.0. Therefore, the diffusion path of CO_2 might be straight as shown in Figure 10(a), leading to a high D_{CO_2}. However, in 1.0 nm pore, the CO_2 diffusion path might be tortuous since the alkyl chain tended to be perpendicular to the wall surface as shown in Figure 10(b). This 'brush like' distribution of alkyl chain may prevent the diffusion of CO_2. To analyze this phenomenon further, the free energy profiles associated with CO_2 permeation for 1.0 and 2.0 nm systems were calculated, as shown in Figure 11. It can be found that there were more obstructions for CO_2 to permeate through IL confined in 1.0 nm pore, since the PMFs for 1.0 nm system fluctuated wildly with large amplitude. The maximum free energy difference of adjacent regions could be as large as 16.5 kJ/mol. On the contrary, the amplitude of CO2 permeation PMFs in 2.0 nm pore was relative small, and the maximum free energy difference was about 5 kJ/mol. In the center of 3.5 nm pore, the preference of $[BMIM]^+$ orientation disappeared and the IL density recovered to the bulk value. Therefore,

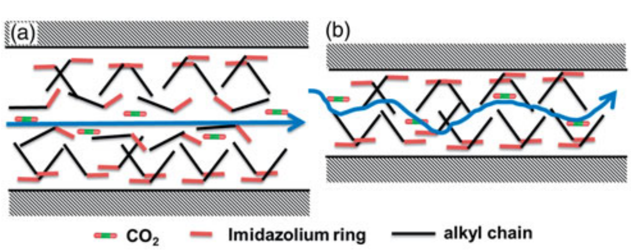

Figure 10. The demonstrations of CO_2 diffusion path in confined ILs. (a) 2.0 nm pore; (b) 1.0 nm pore.

it was reasonable that the D_{CO_2} in BA-3.5 reduced to a small value of $2.7 \times 10^{-10}\,m^2\,s^{-1}$. It was smaller than the estimated D_{CO_2} in bulk [BMIM][Ac] (4.76×10^{-10} $m^2\,s^{-1}$). This was probably caused by the increase of IL viscosity in nanopore.

The S_{CO_2} in BA-1.0, BA-2.0, and BA-3.5 were 698, 274, and 252 $mol\,m^{-3}\,MPa^{-1}$, respectively. The S_{CO_2} in BA-1.0 was extremely high compared with that in other pores. It was understandable because the IL density in the center of 1.0 nm pore was quite small as discussed in "Density profiles" Section. There should be a huge number of free volumes containing CO_2 in this low-density region, leading to a higher S_{CO_2}. The FFV was calculated for the three confined ILs with 0.1 and 0.125 nm probes. As shown in Figure 12, the FFV in confined [BMIM][Ac] decreased as W_p increased, and the variation trends of FFV in BA-2.0 and BA-3.5 were similar. Confined ILs with higher FFV may have more free volumes to absorb more CO_2 molecules.

As shown in Table 2, the estimated P_{CO_2} of [BMIM][Ac] at 400 K are 473, 523, and 207 Barrer for 1.0, 2.0, and 3.5 nm pores, respectively. The experimental P_{CO_2} was 118 Barrer at 293 K for SILMs prepared using [BMIM][Ac] and PVDF (0.22 µm) (Tome et al., 2013). Moreover, the experimental and predicted P_{CO_2} were 10–1000 Barrer at 280–350 K for imidazolium-based ILs (Neves et al., 2010; Chai et al., 2014; Tzialla et al., 2014). Considering the high permeation temperature in this work, the estimated P_{CO_2} were reasonable. Labropoulos et al. (2013) calculated P_{CO_2} of bulk [BMIM][TCM] at 323 K, which was ca. 300 Barrer. According to this, the predicted P_{CO_2} of SILMs prepared with [BMIM][TCM] and silica membrane (1 nm) should be ca. 0.3 Barrer, which was much smaller

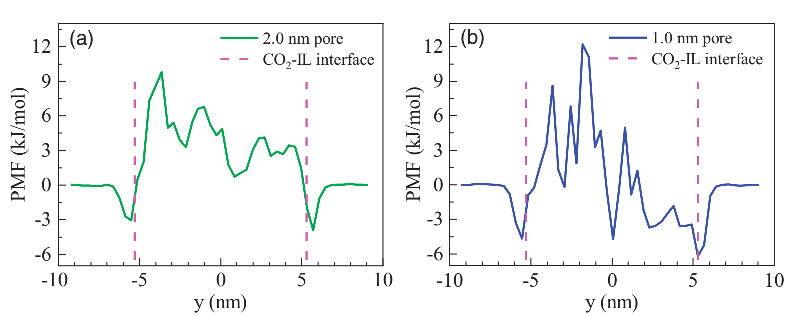

Figure 11. Free energy profiles for CO_2 permeation in confined [BMIM][Ac] (a) 2.0 nm pore; (b) 1.0 nm pore.

than the experimental values (2–12 Barrer). They pointed out that $[BMIM]^+$ might be parallel to the wall surface in the nanopores, leading to a straight diffusion path to facilitate the transport of CO_2. In this work, it was also found that P_{CO_2} of BA-1.0 and BA-2.0 were much higher than that in BA-3.5. Moreover, the high P_{CO_2} in small pore was not only caused by the orientation of alkyl chain (parallel to the wall), but also by the high FFV of ILs confined in 1.0 nm pore.

CO_2 permeation through different confined ILs

Labropoulos et al. (2013) pointed out that the experimental P_{CO_2} of [EMIM][TCM]-based SILMs were close to the predicted values, since the alkyl chain length of $[EMIM]^+$ was small and the orientation of $[EMIM]^+$ might be isotropic. In this section, CO_2 permeation in five ILs ([EMIM][Ac], [BMIM][Ac], [OMIM][Ac], [BMIM][BF4], and [BMIM][PF6]) confined in different pores were discussed to investigate the influence of alkyl chain length and anion type on the CO_2 permeation. The evolutions of CO_2 density profiles in the foregoing confined ILs showed that the alkyl chain length had much stronger influence on the CO_2 permeation than the anion type, as can be seen in Supplementary Figures S7 and S8.

To give a semi-quantitative comparison of CO_2 permeation through different confined ILs, the diffusion coefficient, solubility, and permeability of CO_2 were estimated using the methods in the "Data analysis" Section and the results are listed in Table 3. Given the large calculation quantity of these 12 new systems, no error analysis was performed on these systems. Nevertheless, the errors should be about 15%. The D_{CO_2} in confined [OMIM][Ac] were 21.7, 14.4, and 16.7×10^{-10} $m^2\,s^{-1}$ for 1.0, 2.0, and 3.5 nm pore, respectively. Although in confined [EMIM][Ac], the D_{CO_2} were

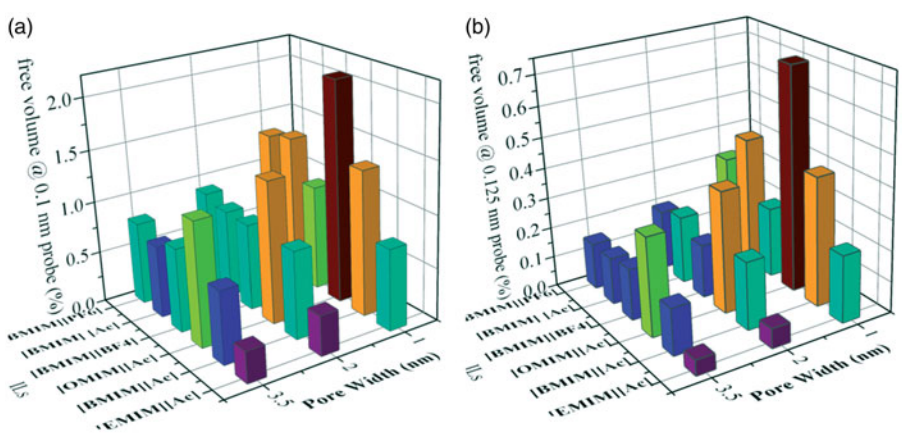

Figure 12. The FFV of confined ILs at 400K for 1.0, 2.0, and 3.5 nm pores. (a) With a 0.1 nm probe; (b) With a 0.125 nm probe.

Table 3. The CO_2 diffusion coefficient, solubility and permeability of five confined ILs for 1.0, 2.0, and 3.5 nm pores.

W_p nm	ILs	Solubility mol m^{-3} MPa^{-1}	Diffusion coefficient 10^{-10} $m^2\,s^{-1}$	Permeability Barrer
1.0	[EMIM][Ac]	445	2.3	299
	[BMIM][Ac]	698	2.4	473
	[OMIM][Ac]	537	21.7	3485
	[BMIM][BF4]	467	2.8	394
	[BMIM][PF6]	722	2.0	435
2.0	[EMIM][Ac]	196	4.1	240
	[BMIM][Ac]	274	6.4	523
	[OMIM][Ac]	510	14.4	2192
	[BMIM][BF4]	249	8.0	593
	[BMIM][PF6]	335	4.7	465
3.5	[EMIM][Ac]	206	2.4	151
	[BMIM][Ac]	252	2.7	207
	[OMIM][Ac]	314	16.7	1561
	[BMIM][BF4]	244	4.1	302
	[BMIM][PF6]	243	3.0	217

2.3, 4.1 and 2.4×10^{-10} $m^2 s^{-1}$ for 1.0, 2.0, and 3.5 nm pores, respectively. The D_{CO_2} in other ILs were 2.0–8.0 $\times 10^{-10}$ $m^2 s^{-1}$ as shown in Table 3. There were few experimental data about the CO_2 diffusion coefficient in confined and bulk ILs, especially at high temperature. Shiflett and Yokozeki (2005) reported the D_{CO_2} in [BMIM][PF_6] and [BMIM][BF_4] at 348.15 K, which were about 2.4 and 3.3×10^{-10} $m^2 s^{-1}$, respectively. According to Scovazzo (2009), the D_{CO_2} in bulk imidazolium-ILs was mainly related to the IL viscosity. Therefore, D_{CO_2} in low-viscosity [BMIM][BF_4] was a little higher than that in [BMIM][PF_6]. The viscosities of ILs used in this work can be found in relevant studies (Harris et al., 2008; Freire et al., 2011; Almeida et al., 2012; Atilhan et al., 2013) and the relationship of D_{CO_2} in the bulk ILs should be [EMIM][Ac] $\approx$ [BMIM][BF_4] > [BMIM][Ac] $\approx$ [BMIM][PF_6] > [OMIM][Ac]. In confined ILs with the same cation [BMIM]$^+$, D_{CO_2} for [BMIM][BF4] were larger than that for [BMIM][Ac] and [BMIM][PF6] in all the pores. It can also be seen from Table 3 that, for [BMIM]$^+$-based confined ILs, the D_{CO_2} in 2.0 nm pore were higher than that in other pores, due to the straight diffusion path for CO_2 in 2.0 nm pore as discussed in "Diffusion coefficient and solubility of CO_2 in confined [BMIM][Ac]" Section.

However, the order of D_{CO_2} in confined ILs with the different cations was [EMIM][Ac] < [BMIM][Ac] < [OMIM][Ac], which was quite different from that in the bulk ILs. Moreover, D_{CO_2} for [OMIM][Ac] was several times larger than that for [EMIM][Ac] and [BMIM][Ac] in all the pores. As discussed above, the parallel orientation of the long alkyl chain of [BMIM]$^+$ in the center of 2.0 nm pore would facilitated the CO_2 diffusion. Therefore, [OMIM]$^+$ may be more favorable to the CO_2 diffusion, since its alkyl chain is longer than [BMIM]$^+$. The density profiles and θ_{IA} distributions of confined [EMIM][Ac] and [OMIM][Ac] are shown in Supplementary Figures S9 and S10. As observed, confined [OMIM][Ac] exhibited much more ordered structure than confined [EMIM][Ac]. The orientation of alkyl chain in confined [OMIM][Ac] was anisotropic even in 3.5 nm pore and tended to be parallel to the wall in the pore center, while it was isotropic for [EMIM][Ac] in the center of 3.5 nm pore.

In 1.0 nm pore, the S_{CO_2} in five confined ILs ranged from 445 to 722 mol m^{-3} MPa^{-1}, larger than those (196–335 mol m^{-3} MPa^{-1}) in 2.0 and 3.5 nm pores. The solubility difference was probably due to the free volume differences in ILs caused by the strong interaction between ILs and walls. As shown in Figure 12, the FFV of ILs confined in 1.0 nm pore was much higher than that in 2.0 and 3.5 nm pores, leading to solubility difference in different pores. In 2.0 and 3.5 nm pores, there was no large difference of CO_2 solubility in all the confined ILs except for [OMIM][Ac]. The CO_2 solubility in confined [OMIM][Ac] was much higher than that in other ILs. This was probably due to the long alkyl chain of [OMIM]$^+$, resulting in a more ordered structure and higher FFV. After comprehensive consideration of D_{CO_2} and S_{CO_2}, the CO_2 permeability, calculated using Equation (3), are also shown in Table 3. The [OMIM][Ac] exhibited the largest CO_2 permeability and it was in a range of 1561–3485 Barrer (3–10 times larger than that of other ILs). The [EMIM][Ac] had the smallest permeability in all condition, which ranged from 151 to 299 Barrer. The difference between the CO_2 permeability of confined [EMIM][Ac] and [OMIM][Ac] was probably caused by the change of the alkyl chain length, which might increase the degree of order in IL structure confined in nanopore.

Conclusions

The study of confined [BMIM][Ac] in γ-Al_2O_3 nanopore (1.0, 2.0, and 3.5 nm) revealed that the confined ILs exhibited a layered structure, with high-density region of imidazolium ring and anion near the wall, and bulk-like region far away from the wall. The alkyl chains near the wall tended to be perpendicular to the wall, while it tended to align with the wall surface in the center of 2.0 nm pore. This parallel orientation of alkyl chain may provide a straight diffusion path for CO_2.

The evolution of CO_2 density profiles showed that CO_2 diffused faster in 2.0 nm pore than in 1.0 and 3.5 nm pores, as CO_2 reached a relative even distribution in only ca. 15 ns. The diffusion coefficient and solubility of CO_2 in different ILs ([BMIM][Ac], [EMIM][Ac], [OMIM][Ac], [BMIM][BF_4], and [BMIM][PF_6]) confined in 1.0, 2.0, and 3.5 nm pores were estimated and gave a semi-quantitative indication of the CO_2 permeation changes under confinement. It was found that the relationship of D_{CO_2} in confined ILs with the same $[BMIM]^+$ was consistent with that in the bulk ILs, and D_{CO_2} in 2.0 nm pore were higher than that in 1.0 and 3.5 nm pores, due to the straight diffusion path.

However, the D_{CO_2} in confined [EMIM][Ac], [BMIM][Ac], and [OMIM][Ac] were 2.3–4.1, 2.4–6.4, and $14.4–21.7 \times 10^{-10}$ $m^2\,s^{-1}$, respectively. This order was opposite to that in the bulk ILs, since the longer alkyl chain resulted in a more ordered structure, facilitating CO_2 diffusion. In addition, the S_{CO_2} in five confined ILs were 445–722 $mol\,m^{-3}\,MPa^{-1}$ in 1.0 nm pore, which were larger than those (196–335 $mol\,m^{-3}\,MPa^{-1}$) in 2.0 and 3.5 nm pores, which may be caused by the higher FFV of ILs in 1.0 nm pore. Both parallel orientation of alkyl chain and high FFV of ILs in nanopores could increase the CO_2 permeability in confined ILs.

Nomenclature

Latin

D	self-diffusion coefficient ($m^2\,s^{-1}$)
W_p	pore width of the γ-Al_2O_3 pore (nm)
H	Henry constant of gas in ILs (MPa)
L	the dimension of simulation box (nm)
m_t	the weight of gas in ILs at t (kg)
m_∞	the weight of gas in ILs at equilibrium (kg)
P	gas permeability of ILs (Barrer)
S	gas solubility of ILs ($mol\,m^{-3}\,MPa^{-1}$)
V_m^{ILs}	molar volume of ILs ($m^3\,mol^{-1}$)

Greek letters

θ_{XX}	the angle between Z direction and vector XX
φ_{XX}	the angle between Y direction and vector XX

Abbreviations

IL	ionic liquid
ILs	ionic liquids
EMIM	1-ethyl-3-methyl-imidazolium
BMIM	1-butyl-3-methyl-imidazolium
HMIM	1-hexyl-3-methyl-imidazolium
OMIM	1-octyl-3-methyl-imidazolium
RMIM	alkyl-methyl-imidazolium
BF_4	tetrafluoroborate
PF_6	hexafluorophosphate
Tf_2N	bis(trifluoromethylsulfonyl)imide
TCM	tricyanomethanide
Ac	acetate

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